

DEVARSH PATEL

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EDUCATION

Master of Science in Information Management | University of Washington, USA **GPA: 4.0 | Jun 2025**

Relevant Courses: Data Analytics (Statistics), Business Analysis, Business Intelligence, Product Design & Management

Bachelor of Technology in Information Technology | NMIMS University, India **GPA: 3.8 | Aug 2023**

Relevant Courses: Data Analysis, Data Structures and Algorithms, Relational Database Management System, Cloud Computing

PROFESSIONAL EXPERIENCE

Data Analytics Engineer | Hitachi Americas (Capstone) **Dec 2024 – Present**

- **Real Time Analytics** - Developing analytics dashboard to analyse chatbot's engagement, query resolution rates and productivity impact by leveraging Databricks enabling data driven decision making
- **Big Data & Cloud** – Implemented an Apache Kafka and PySpark pipeline on Docker to transform structured and unstructured data into Databricks data lake, boosting retrieval efficiency for prompts on Buddy AI chatbot
- **Problem Solving** - Engineered multimodal Retrieval-Augmented Generation (RAG) based LLM chatbot, cutting employee query resolution time from 6 days to 1 day
- **Collaboration** – Analysed the tool, company and client side requirements for the enterprise knowledge Buddy AI chatbot

Data Analyst | University of Washington, Seattle, USA **Oct 2023 – Nov 2024**

- **Requirement Analysis** – Collaborated with stakeholders to gather reporting needs to define technical KPI's for reports
- **Exploratory Data Analysis** - Conducted EDA & statistical analysis using R for reports on 20 KPIs uncovering key trends
- **Advanced Analytics** - Built Power BI dashboards for data visualization providing actionable insights, speeding up informed decision making by 30% for strategic changes in college
- **ETL Pipeline** - Orchestrated ETL workflows using Airflow and SQL for data transformation enabling efficient analysis of college performance

Data and Analytics Engineer Intern | OTC Markets Group, New York City, USA **Jun 2024 – Aug 2024**

- **Cost Optimization** – Designed & implemented in-house automated data dictionary tool, saving \$180K annually eliminating its reliance on third-party subscription costs like Alation
- **Database Architecture** - Architected a centralized financial metadata repository using Python and SQL, for AWS PostgreSQL, MariaDB, and Informix databases accelerating SEC reporting by 75%
- **DevOps & Infrastructure** - Configured Red Hat Linux virtual machine using bash for cron jobs to update data automatically reducing manual intervention by 60% ensuring robust environments

Data Analyst Intern | KPMG (Government & Public Sector), Noida, India **May 2022 – Jul 2022**

- **Dashboarding** - Developed a voice-enabled Tableau dashboard using Natural Language Processing (NLP), SpeechRecognition, & pyttsx3 reducing filter time by 50% for India's Health Ministry through cross-functional collaboration
- **Data Processing** - Streamlined data processing with GCP Dataflow and BigQuery, automated excel generation and WhatsApp delivery of daily reports to 40 members using Python, reducing manual workload by saving 2 hours daily
- **Query Optimization** - Optimized SQL queries for 200K+ COVID data; applied z-test and chi-squared test for data validation

ACADEMIC PROJECTS AND PAPERS

Retail Store Data Business Intelligence System **Jan 2024 – May 2024**

- **Revenue Analytics** - Built a PySpark based ELT pipeline integrating 150K+ records into Snowflake data warehouse using SQL for EDA, enabling sales & revenue analytics
- **Dashboarding & Forecasting** - Forecasted sales using XGBoost regression (Scikit-learn) and built Tableau dashboards, improving targeted marketing by 15% driving business decisions such as profit allocation, store closures for GTM strategies
- **Dimensional Modeling** – Conducted dimensional data analysis followed by normalization and star schema design on Snowflake for dimensional and fact tables

Multilingual Hinglish to English Translation and Classification Techniques [Research Paper] **Jul 2022 – Mar 2023**

- **Machine Learning** - Developed a novel recurrent neural network (RNN) LSTM model for token classification of Hindi or English achieving 91% accuracy in transliteration of Hinglish (Hindi + English) and translation of English tokens
- **Natural Language Processing** - Improved translation of Hinglish text to English equivalent by 15% by leveraging machine learning, python, and NLP techniques

SKILLS

- **Languages:** Python, R, SQL, NoSQL
- **Databases:** AWS (Postgres, DynamoDB), GCP (BigQuery), Snowflake, Informix, MariaDB
- **BI Tools:** Tableau, PowerBI, Looker
- **Big Data & DevOps:** Apache (Kafka, Spark, Airflow), Docker
- **Concepts:** Statistics, Relational Databases, Data Warehousing, CI/CD, OLAP, OLTP, Machine Learning, Generative AI